Logarithms

1

a $\log_5 25 = 2$

b $\log_3 81 = x$

c $\log_3 3 = 1$

d $\log_2 1 = 0$

e $\log_{4.4} 1 = 0$

f $\log_{25} 125 = 1.5$

g $\log_4 32 = \frac{5}{2}$

h $\log_4 0.0625 = -2$

i $\log_4 \frac{1}{64} = -3$

j $\log_x 64 = 1.5$

k $\log_4 9 = m$

l $\log_5 q = m$

m $\log_h w = q$

n $\log_m n = j$

2

a $\log_{10} 100 = 2$

b $\log_{10} \left(\frac{1}{100}\right) = -2$

c $\log_{10} \left(\sqrt{10}\right) = \frac{1}{2}$

d $\log_{10} \left(\frac{1}{\sqrt{10}}\right) = -\frac{1}{2}$

3

a $4^2 = 16$

b $5^1 = 5$

c $8^0 = 1$

d $2^{-3} = 0.125$

e $3^{-1} = \frac{1}{3}$

f $(5.8)^2 = 33.64$

g $\left(\frac{1}{3}\right)^{-2} = 9$

h $x^5 = 32$

i $8^6 = x$

j $g^3 = 5$

k $7^{40} = m$

l $y^u = k$

m $r^y = p$

4

a $10^4 = 10\,000$

b $10^{\frac{1}{2}} = \sqrt{10}$

c $10^{-4} = \frac{1}{10\,000}$

d $10^{-\frac{1}{2}} = \frac{1}{\sqrt{10}}$

5 True

Evaluating logarithms

6

a

i $x = \log_{10} 820$

ii 2.91

b

i $x = \log_{10} 0.0002$

ii -3.70

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c

i $x = \log_{10} \frac{13}{50}$ ii -0.59

d

i $x = \log_{10} 14\,000$ ii 4.15

7

a

i 10^6 ii 6

b

i 10^{-6} ii -6

c

i 10^0 ii 0

d

i 10^{-1} ii -1

8

a 6 **b** -1 **c** -2 **d** 0

e 2 **f** -2 **g** -2 **h** $\frac{1}{3}$

i -4

9

a

i -1 ii -1 iii -1 iv -1

b 1

10

a 1 **b** 2 **c** Between 1 and 2

11

a

i between 3 and 4 ii 3.9490

b

i between -2 and -1 ii -1.2757

c

i between 2 and 3 ii 2.7275

d

i between 0 and -1 ii -0.1101

12

a 1.60 **b** 3.89 **c** -2.24 **d** 0.90

e 1.89 **f** 1.67 **g** -2.81 **h** -2.25

i -4.52 **j** 10.22 **k** 2.07 **l** -6.50

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13 316.23

14 0.063

15 1

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